

SPECTRAVIDEO USERS GROUP

WELLINGTON

NEWSLETTER

Vol 4 - Number 6
SEPTEMBER-OCTOBER 1988

EDITOR: John Hodgson
(00 64) 4 367550

Meetings are held on the third Monday of the month at 7.30 pm in the Staff Room of Newlands Intermediate, Bracken Road, Newlands, Wellington. Attendance currently averages 10-15 with ideas, information, problems and matters of interest being shared.

COMMITTEE MEMBERS

Chairman	Jim Oliver	ph 665 205
Secretary/Treasurer	Ross Fletcher	ph 788 454
Librarian	Scotty Baxter	ph 784 619

Public Domain BASIC	Jim Oliver
Public Domain CP/M	Ross Fletcher

NEWSLETTER CONTRIBUTIONS TO: J. Hodgson, 38 Gloaming Hill
Onepoto
WELLINGTON

C O N T E N T S

:	:	:
:	:	:
:	Editorial	page 2
:	:	:
:	Letters to the Ed	page 2
:	BRAIN STRAIN	page 4
:	Art Program - Michael Warwick	page 5
:	Wordprocessing - Other Languages	page 8
:	80 Column Screen Print Dump	page 9
:	Print80.BAS - Correct Version	page 10
:	Conix System Update	page 10
:	New Public Domain CP/M Files	page 11
:	Screen Print from CP/M	page 13
:	Public Domain Basic Files List	page 16
:	SVI BASIC TOKENS - numerical order	page 20
:	:	:
:	:	:

From your Editor,

The Editors Prize of \$20 this issue goes to Michael Warwick for his Art program.

Keep those subs coming in and remind your friends too.

LETTERS DEPARTMENT

..from Dennis and Jenney Browne, Tairua, NZ

Dear Everybody,

As usual you are all still doing great things - the magazine is always a shot in the arm for me.

After reading the letter 'SV in Hong Kong' by Henry Chiu, I'd like to say that if anyone is collecting orders for a production run would you put my name down for an 80 col card for my 605 expander and if you need another name for anything else please think of me as I'll certainly consider it.

I'm trying not to feel so guilty about being rather useless in the programing department - but thats not to say I've given up - as they say, ONE DAY! and then you can print it with bells on.

318s and 328s would be dead if it weren't for you all and we (the greenhorns) out here would never have known how versatile our little SVI 8 bits are.

Please be assured that all the work you do is really appreciated in this tiny office on the Coromandel Penninsula.

Jenney Browne.

Right on Jenney. Thanks from 'us all' for the encouragement - and keep plugging away at that programming. Whatever is learnt on the SVI in the way of good programming and languages is all valid for other machines!

...ed

WANTED TO BUY

SV806 80 COLUMN CARD

- contact Paul Earl
7 Bellevue Street
Roslyn
DUNEDIN

Anyone who can help Jeff please write to him direct

...ed

24-07-1988

Spectravideo Users Group - Wellington,
C.-The Editor, Mr. John Hodgson,
38 Gloaming Hill
Onepoto.
WELLINGTON.

Dear John,

Jim Oliver referred me to you regarding you would be able to send me the very first 13 issues (prior to September 1985) of your Newsletter, so I have the complete series.
Jim has set the price at \$ 3.00 a copy, so please find enclosed a sum of \$39.00 + \$18.00 to renew my Subscription which falls due 10-08-1988 (Membership Number 2034) ; please advise Ross Fletcher that I have renewed my Sub. : the cheque total is for \$57.00 .

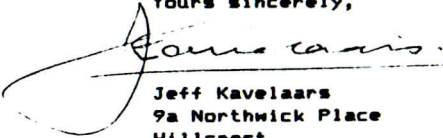
Congratulation with your last Newsletter : it was excellent.

I also subscribe to the O.L.S. (On Line Support) Team Newsletter, based in Christchurch
They were not able to help me with a problem I encountered and I am wondering if you could throw some light on the subject.
The question I put to them was as follows:

"I'm having a small problem with my Wordprocessor, called SuperWriter which run under CP/M 2.22 operating system. I have a Spectravideo 328 with 2 disc drives and an 80 column card etc. I want to include Keyed sections ("Boilerplates") from a 'Keyed file' into the document I am writing.
To make those Key words or phrases I have to enter a formfeed character , enter a Key word or phrase to identify the text and after the Return either type or include the 'boilerplate' text. In my SuperWriter manual it says : Refer to the installation instructions how to enter the Form Feed (FF) character, however I can't find it in my CP/M manual.
For my printer if I want to include the FF character you type CTRL L, but if this was the same for the CP/M operating system how could you SAVE CTRL L in a Keyed file for later use ? "

Could YOU help please?

Yours sincerely,


Jeff Kavelaars
9a Northwick Place
Hillcrest
Auckland 10
Ph. (09) 4182754

CORRECTION

CORRECTION TO 80 COLUMN PEEK POKE INSTRUCTIONS BY JIM OLIVER

In the May June issue of the newsletter we (I) said on page 8 in the basic program ,line 110 this :

```
110 CMDPEEKX,Y:LPRINTCHR$(PEEK(54545))
```

If you were assembling the latest version of the various 80 columns peek poke routines called CMDPEEK/POKE/LINE on page 3 then the number after peek should be 54543 (&HD0F). NOTE the three at the end and not 5. Only the FIRST version of 80 col peek/poke on page 15 works using 54545. (&HD511)

Bob advises anybody using these files to now only use the latest version on page 3.

If Bob was'nt so fast at coming up with new improved versions with boxes added we (I) would'nt get this sort of thing wrong.

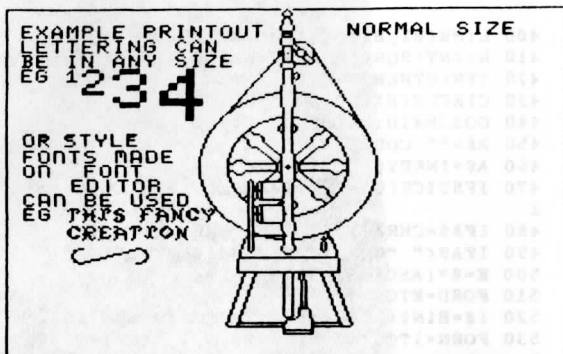
Bob's little demo program on page 7 is VERY impressive as those who have seen it will testify.

BRAIN STRAIN DEPARTMENT

C'mon - where are all these computer whizz kids? We are still waiting for an answer to the No 2 Teaser!

```
10 REM BIN RD by D.Kelly SVUG DEC 85
20 REM modified for MSX by G.Clark April 1988
30 AD=&HD000
40 READ D$
50 IF D$="*" THEN 210
60 POKE AD,VAL("&H"+D$)
70 AD=AD+1
80 GOTO 40
90 DATA 0E,D0
100 DATA CD,B8,70
110 DATA CD,15,71
120 DATA CD,E9,72
130 DATA CD,0B,70
140 DATA 22,30,D0
150 DATA CD,0B,70
160 DATA 22,32,D0
170 DATA CD,0B,70
180 DATA 22,34,D0
190 DATA C9
200 DATA *
210 FOR F=&HF866 TO &HF86D
220 POKE F,32
230 NEXT F
240 DEFUSR=&HD000
250 A$=USR(0)
260 PRINT
270 AD=&HD030
280 PRINT"START: ";
290 GOSUB 360
300 PRINT"END : ";
310 GOSUB 360
320 PRINT"EXEC : ";
330 GOSUB 360
340 PRINT
350 END
360 PRINT HEX$(PEEK(AD)
+256+PEEK(AD+1))
370 AD=AD+2
380 RETURN
```

ART PROGRAM - MICHAEL HARNICK



```

10 DEFINT A-Y:DIMP(767):GOSUB600
20 DIMC(10,10):M=1:SP=1
30 COLOR1,15,1:SCREEN1:GOSUB 360
40 GOSUB 360:X=STICK(0):Y=STRIG(0)
50 X=STICK(0):Y=STRIG(0)
60 A$=INKEY$
70 IFA$="+ "THENC=C+1
80 IFA$="- "THENC=C-1
90 IFA$="S"THENSARE"1:SCREEN",S
100 IFA$="L"THENLOAD"1:SCREEN":GOSUB360
110 IFA$=","THENSP=SP+1:IFSP>5THENSP=1
120 IFA$="."THENBX=BX:BY=SY:PSET(BX,BY),C
130 IFVAL(A$)<LORVAL(A$)>8THEN160
140 ONVAL(A$)GOSUB590,580,450,440,410,400,380,390
150 GOTO210
160 IFA$="9"THENPAINT(SX,SY),C
170 IFA$="*"THENPRESET(SX,SY)
180 IFA$="/"THENCLS
190 IFA$=CHR$(16)THENGOSUB610
200 IFA$=CHR$(27)THENRUN"1:MENU"
210 IFC=16THENC=0ELSEIFC=-1THENC=15ELSEC=C
220 IFX=1THENSY=SY-SP
230 IFX=2THENSY=SY-SP:SY=SY+SP
240 IFX=3THENSX=SY+SP
250 IFX=4THENSX=SY+SP:SY=SY+SP
260 IFX=7THENSX=SY-SP
270 IFX=5THENSY=SY+SP
280 IFX=6THENSX=SY-SP:SY=SY+SP
290 IFX=8THENSX=SY-SP:SY=SY-SP
300 IFSX>256THENSX=0
310 IFSY>192THENSY=0
320 IFSX<0THENSX=256
330 IFSY<0THENSY=192
340 IFY<0THENPSET(SX,SY),C
350 PUTSPRITE1,(SX-2,SY-3),C,1:GOTO50
360 SPRITE$(1)=CHR$(32)+CHR$(32)+CHR$(216)+CHR$(32)+CHR$(32):SX=128:SY=96:C=1:RE
TURN
370 BX=X:BY=Y:PSET(BX,BY),C:RETURN
380 LINE(BX,BY)-(SX,SY),C,BF:RETURN
390 LINE(BX,BY)-(SX,SY),C,B:RETURN

```

```

400 LINE(BX,BY)-(SX,SY),C:RETURN
410 R=INT((SQR((ABS(SX-BX)^2)+(ABS(SY-BY)^2))+.5):REM( THANKS TO PYTHAGORUS! )
420 IFR<OTHERNR=R*-1
430 CIRCLE(BX,BY),R,C:RETURN
440 GOSUB410:PAINT(BX,BY),C:RETURN
450 A$="":COLORC
460 AS=INKEY$
470 IFSTICK(0)=0THENR=0ELSEIFSTICK(0)=5THENM=1ELSEIFSTICK(0)=1ANDR=0THENM=M+1:R=
1
480 IFA$=CHR$(13)THENRETURN
490 IFA$<" "ORAS>" "THEN460
500 E=8*(ASC(A$)-32)
510 FORD=ETOE+7
520 I$=BIN$(P(D)):I$=STRING$(8-LEN(I$),"0")+I$
530 FORN=1TO6:B$=MID$(I$,N,1):IFB$="1"THENLINE(SX,SY)-(SX+M-1,SY+M-1),C,BF
540 SX=SX+M:NEXT SX=SX-6*M:SY=SY+M:NEXT SY=SY-8*M:SY=SY+7*M
550 IFSX>255-7*MTHENSX=2:SY=SY+8*M
560 IFSY>191-5*MTHENRETURN
570 GOTO460
580 GET(SX,SY)-(SX+9,SY+9),C:PUT(SX,SY),C,PSET:PUT(0,0),C,PSET:PUT(SX,SY),C,PS
ET:RETURN
590 PUT(SX,SY),C,PSET:RETURN
600 FORA=0TO767:P(A)=VPEEK(A+2048):NEXT:RETURN
610 QS=INPUT$(1):IFQS="1"THEN620ELSEIFQS="2"THEN720ELSE610
620 OPEN"LPT:"FOROUTPUTAS#1:PRINT#1,CHR$(27)+"3"+CHR$(24);
630 FORZ=0TO191STEP8
640 PRINT#1,CHR$(27)+"*"+CHR$(0)+CHR$(0)+CHR$(1);
650 FORRX=0TO255:D=0:P=8
660 FORRY=ZTOZ+7:RA=POINT(RX,RY):P=P-1
670 IFRA=15THENRA=0ELSERA=2^P
680 D=D+RA:NEXT:PUTSPRITE1,(RX-2,Z),C,1
690 IFD=9THEND=13ELSEIFD=27THEND=25
700 PRINT#1,CHR$(D);:NEXT
710 PRINT#1,CHR$(27)+CHR$(10):NEXT:RETURN
720 OPEN"LPT:"FOROUTPUTAS#1:PRINT#1,CHR$(27)+"3"+CHR$(24);
730 FORZ=0TO191STEP4:PRINT#1,CHR$(27)+"*"+CHR$(0)+CHR$(0)+CHR$(2);
740 FORRX=0TO255:D=0:P=7:FORRY=ZTOZ+3:RA=POINT(RX,RY):P=P-1
750 IFRA=15THENRA=0ELSERA=2^P
760 D=D+RA+(RA*2):P=P-1:NEXT:PUTSPRITE1,(RX-2,Z-1),C,1
770 IFD=9THEND=13ELSEIFD=27THEND=25
780 PRINT#1,CHR$(D);CHR$(D);:NEXT:PRINT#1,CHR$(27)+CHR$(10):NEXT:RETURN

```

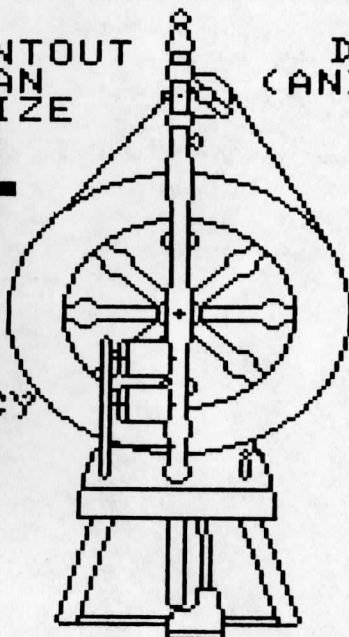
ARTIST HELP MENU

KEY/S	ACTION	KEY/S	ACTION
[+]	MOVES CROSS RIGHT	[9]	PAINTS IN AREA FROM POSITION OF CROSS
[+]	MOVES CROSS LEFT	[8]	EMPTY BOX FROM POINT IN MEMORY TO CROSS POSITION
[+]	MOVES CROSS UP	[7]	FILLED BOX (SEE [8])
[+]	MOVES CROSS DOWN	[6]	LINE FROM POINT IN MEMORY TO CROSS POSITION
[/]	CLEARs THE SCREEN	[5]	CIRCLE WITH POINT IN MEMORY AS CENTRE AND CROSS POSITION AS OUTSIDE
[SPACE]	LIGHTS UP DOT IN CENTRE OF CROSS TO PRESET CROSS COLOUR.	[4]	AS WITH [5] EXCEPT WILL FILL
[*]	RESETS DOT IN CENTRE OF CROSS TO BACKGROUND COLOUR (WHITE).	[3]	SET TYPE MODE-[+]INCREASES SIZE [0]DEFAULT SIZE
[+]	CHANGES CROSS COLOUR NEXT ONE UP THE SCALE (SEE BELOW)	[2]	PHOTOGRAPHS AN AREA (10x10)
[-]	CHANGES CROSS COLOUR DOWN ONE THE COLOURS ARE:	[1]	PRINTS PHOTOGRAPH TO SCREEN
	0) TRANSPARENT	[L]	LOAD SCREEN FROM DISK
	1) BLACK	[S]	SAVES SCREEN TO DISK
	2-14) SHADES BETWEEN BACK & WHITE	[CTRL]+[P]	FOLLOWED BY
[.]	SETS POINT IN MEMORY (DEFAULT-TOP LEFT HAND CORNER)	[1]	SCREEN TO PRINTER-NORMAL SIZE
[,]	SPEEDS UP OR SLOWS CROSS	[2]	SCREEN TO PRINTER-DOUBLE SIZE

EXAMPLE PRINTOUT
LETTERING CAN
BE IN ANY SIZE
EG 1234

DOUBLE SIZE
(AND DOUBLE TIME)

OR STYLE
FONTS MADE
ON FONT
EDITOR
CAN BE USED
EG THIS FANCY
CREATION



WORDPROCESSING IN OTHER EUROPEAN LANGUAGES

Most printers provide several character sets. They are intended to provide for printing the accented characters in certain European languages. Some time ago Bill Fergusson wrote asking how to use the extra characters needed for word-processing in German. Other readers may also have experience with the frustrations of using Wordstar for languages other than English. Wordstar is far from ideal when it comes to using any accented letters. There are two main problems:

(1) Wordstar uses character codes 128 to 255 for its own purposes (to mark the end of words in document mode) and so you can't tell it, for example to use ASCII 142 for Ä. In fact you can't easily use any high-bit characters;

(2) the 80-column card has only one character set and does not allow for fonts to be redesigned. This problem is insurmountable as far as I know, but what follows is a work around for this problem.

My printer (a Star Gemini 10X) has provision for German characters (and several other languages). There are two ways to set them up:

(1) by setting the DIP switches for German (1.6 on, 1.7 off, 1.8 on. Similar but different DIP settings are used for other printers.

(2) by software selection, eg. LPRINT CHR\$(27)CHR\$(55)CHR\$(2), ie. ESC, "7", 2. In order to get Wordstar to send these characters to the printer you will need to patch the program, for example using WINSTALLS, so that, one of the user redefinable commands, say CTRL F W is ESC "7" 2, to change to German, and CTRL F E, say, is ESC "7" 0, to get back to the USA set, which is the default (or whatever the codes are on your printer for the same functions).

Once you have redefined this pair of commands you can use them in a Wordstar document. Use ^PW to toggle into German mode and instead of the [\] { : } ~ which appear on the screen, the letters Ä Ö Ü ä ö ü Æ should be printed by the printer. If you want to use any of these brackets etc in the same text just toggle back to the default set by inserting ^PE in the text. All the unlauted characters will appear as brackets etc in the text, which makes the text hard to read, but for occasional use it is tolerable.

D E Rogers

80 COLUMN SCREEN PRINT DUMP By Jim Oliver

Elsewhere in this issue you will find an assembler program written by Bob Curwood from the Christchurch Users Group. It has occurred to me that many people may not be able to make use of this since they either don't have an assembler or like me they simply don't know what it's all about !

If you fall into either of these two categories read on, help is at hand. I have figured out enough using a marvellous little program called ZEN to try out Bob's routine. It is MARVELOUS!

Do you remember the 40 column print screen routine in the Jan/Feb issue ? Well this routine will allow you to dump the 80 column screen direct to your printer (with some constraints, read Bob's notes at the end of his assembler program).

I've converted all that assembler language to data which will allow you create the machine code disc file from BASIC without having to own a copy of ZEN or understanding whats going on. Isn't that jolly decent of me ?

Type the following program in carefully and then save it to disc THEN run it. You should end up with a file on disc which can be activated by typing BLOAD "1:PRNT80.M/C",R. After that the rest is up to you--- hint--- fill the 80 column screen with anything, then hit the PRINT key ! If you have a more or less standard printer then it'll work for you. If it doesn't work then drop me a line.

Since this machine code is located at &H9000 to &H9064 I assume that if your BASIC program is larger than about &H9000 minus &HB001 bytes long then you will wipe this great little routine out of existence. It can be relocated in memory but you will have to work that one out yourself.

p.s Bob is going to convert this for use under CP/M so that you can dump Dbase II or Wordstar or any screen directly to your printer, so watch this space friends !

```

10 FOR I=&H9000 TO &H9064: READ A$:POKE I,VAL("&H"+A$):E=E+VAL("&H"+A$):NEXT
30 IF E <> 11194 THEN GOTO 150
40 '
50 DATA 21,C,90,22,7A,FE,3E,C3,32,79,FE,C9,3E,8,D3,96,DB,99,E6,20,C0,DB
60 '
70 DATA 12,E6,1,C0,21,0,F0,11,30,F7,3E,A,CD,15,39,F3,3E,FF,D3,58,4E,3C
80 '
90 DATA D3,58,23,3E,6,D3,96,DB,99,E6,2,20,A,3E,2,D3,96,DB,99,E6,8,28,1E
100 '
110 DATA 79,FE,5F,38,8,FE,BE,30,8,C6,40,18,6,C6,20,18,2,0,0,CD,15,39,E7
120 '
130 DATA 20,C9,3E,A,CD,15,39,CD,BE,40,C9
140 BSAVE"1:PRNT80.M/C",&H9000,&H9064:PRINT"CODE SAVED TO DISC":END
150 PRINT"ERROR IN DATA --- CHECK TYPING":END

```

THE FOLLOWING IS THE CORRECT WORKING COPY OF PRINT80.BAS - SEE
PREVIOUS ISSUES - Jim Oliver

```
10 FOR I=&H9000 TO &H9065: READ A$:POKE I,VAL("&H"+A$):E=E+VAL("&H"+A$):NEXT
30 IF E <> 11194 THEN GOTO 160
40
50 DATA 21,C,90,22,7A,FE,3E,C3,32,79,FE,C9,3E,8,D3,96,DB,99,E6,20,C0,DB
60
70 DATA 12,E6,1,C0,21,0,F0,11,30,F7,3E,A,CD,15,39,F3,3E,FF,D3,58,4E,3C
80
90 DATA D3,58,23,3E,6,D3,96,DB,99,E6,2,20,A,3E,2,D3,96,DB,99,E6,8,28,1E
100
110 DATA 79,FE,5F,38,8,FE,BE,30,8,C6,40,18,6,C6,20,18,2,0,0,CD,15,39,E7
120
130 DATA 20,C9,3E,A,CD,15,39,CD,BE,40,C9,00
140
150 BSAVE"1:PRINT80.M/C",&H9000,&H9065:PRINT"CODE SAVED TO DISC":END
160 PRINT"ERROR IN DATA --- CHECK TYPING"
170 END
```

ConIX OPERATING SYSTEM UPDATE By Jim Oliver

I continue to experiment with ConIX and have found that it automatically installs perfectly under Steve McNamee's SV Customised CP/M. All of the convenient features available under SV Custom CP/M that I have tried work perfectly under ConIX. Most of the normal ConIX commands work OK also; making the combination of SV CP/M and ConIX a real winner if you really want to make use of CP/M.

Elsewhere in this issue you will find a PD CP/M listing of new files available. Like a lot of others I dislike unnecessary typing and so when it came time to actually type up the article I had two alternatives, either list the disc directories on the printer (using ^P then DIR) and then copy this through the keyboard to a Wordstar file or find some automatic way of reading the directory tracks directly to a disc.

I tried the CP/M cataloguing system but found this did not give the size of files in the listing, i.e no good, (but excellent to keep a catalogue of your CP/M discs). I then hit on the idea of using the ConIX OUTPUT REDIRECTION feature.

After installing the ConIX to a SV Custom disc and checking it worked it was a very easy matter to type the following from the ConIX prompt D B: > DIRLIST1. This runs the file D.COM (a sorted directory listing of files and sizes) on the B drive (from A) and then directs the output of this action to a new file called DIRLIST1 on the A drive (instead of to the screen). Problem solved !

If you want to find out more about ConIX please contact me.

***** NEW PUBLIC DOMAIN CP/M FILES *****

We received recently from (our valued regular contributor) Bob Curwood some new CP/M files to add to the PD discs already in our library. I have now checked and sorted these out and placed them on discs in what I think is the most convenient way. That's to say you should, (NO guarantees) get all the files you want on any particular subject on one disc. Have you noticed that with CP/M discs you generally need them ALL to get all the files you want?

I have unsqueezed them and copied them onto discs using the SV CUSTOM FORMAT by Steve McNamee, Australia. I have done this to eliminate any unsqueezing problems you may have and also you will remember the custom format discs hold 182 K of data in place of the normal 155 K of data. This is still a bit worse than storing the files in squeezed format on standard discs but it does eliminate the sometimes troublesome unsqueezing. As well as that it will be a good chance for a few of you to see how good the custom format really is!

DISC NO 1A

-BCCMPD 01A	0K	:	ADVENTUR	DOC	2K	:	ATAB	DAT	2K	:	TRAVEL	DAT	6K
AD	COM	40K	:	ADVENTUR	MSG	64K	:	BUG	BAS	2K			
ADVENTUR	COM	22K	:	ADVENTUR	WRK	42K	:	TOKEN	BAS	2K			

10 Files (13 entries) using 182K of total 182K capacity
51 Directory entries and 0K Bytes available on Drive B:

DISC NO 1B

-BCCMPD 01B	0K	:	COMPARE	COM	2K	:	DISPLAY	DOC	4K	:	SUPERSUB	DOC	16K
BC	0K	:	COMPARE	DOC	2K	:	FMAP	COM	4K	:	SYS215	COM	8K
BSORT	COM	4K	:	COPYFILE	ASM	4K	:	FMAP3	COM	2K	:	TOKEN	BAS
BSORT	NOT	4K	:	COUNT	ASM	6K	:	KTAB	DAT	2K	:	UCAT	COM
CAT	COM	2K	:	CPMPower	COM	12K	:	LIST	ASM	10K	:	UCAT2	COM
CAT1	COM	2K	:	CPMPower	OVR	6K	:	MAST	CAT	2K	:	UNSPool	COM
CAT2	COM	2K	:	CRC	COM	4K	:	OPEN	ASM	4K	:	UNSPool	DOC
CATALOG	DOC	8K	:	CRC	DOC	2K	:	QCAT	COM	2K	:	USQ	COM
CHEK	COM	4K	:	DIRCHK	COM	2K	:	REVIEW	COM	6K	:	WCOUNT	COM
CLOSE	ASM	4K	:	DIRCHK	DOC	2K	:	SCRAMBLE	MSG	4K	:	WCOUNT	DOC
COMMON	DAT	4K	:	DISPLAY	COM	2K	:	SUPERSUB	COM	4K			

43 Files (43 entries) using 178K of total 182K capacity
21 Directory entries and 4K Bytes available on Drive B:

DISC 2A

-BCCMPD 02A	0K	:	BYE	DOC	2K	:	MEX	COM	24K	:	TUTL	PAS	14K
ALLOC	COM	2K	:	BYEZCPR	DOC	10K	:	MODEMNS	TXT	4K	:	UTE	HLP
BYE	ASM	64K	:	DISK	DOC	2K	:	SD	COM	4K			
BYE	COM	4K	:	HAYES	TXT	14K	:	TUTL	COM	28K			

14 Files (15 entries) using 180K of total 182K capacity
49 Directory entries and 2K Bytes available on Drive B:

DISC 2B

-BCCMPMD 02B 0K : DU-V86 COM 8K : MACASM COM 8K : XERA ASM 14K
 DOWNLOAD COM 2K : DU-V86 DOC 14K : ROFF4 COM 32K
 DU-V86 ASM 56K : MAC COM 12K : ROFF4 DOC 34K
 10 Files (12 entries) using 180K of total 182K capacity
 52 Directory entries and 2K Bytes available on Drive B:

DISC 3A

-BCCMPMD 03A 0K : EX14 DOC 6K : LISTIT INC 12K : SVINST11 COM 14K
 E COM 20K : EXPRESS OVL 6K : LISTT COM 2K : TERM DAT 14K
 ECONFIG1 COM 12K : FINDBD54 COM 2K : LISTT DOC 4K : U-SVDO COM 8K
 EX COM 4K : FINDBD54 DOC 2K : LISTT PAS 14K : VTERM DAT 4K
 EX14 COM 4K : FIX19 COM 36K : SVDO COM 8K
 19 Files (20 entries) using 172K of total 182K capacity
 44 Directory entries and 10K Bytes available on Drive B:

DISC 3B

-BCCMPMD 03B 0K : DBUGABST DOC 28K : INSTRUCTS DOC 12K : UNLOAD COM 2K
 ATAB DAT 2K : DBUGINST DOC 10K : KTAB DAT 2K : VDO MAN 22K
 COMMON DAT 4K : DEBUG COM 4K : MLOAD21 COM 4K : VFILER COM 8K
 CPMDEBUG ASM 34K : DEBUG Z80 32K : SYMSG BB 12K : VTERM DAT 4K
 16 Files (18 entries) using 180K of total 182K capacity
 46 Directory entries and 2K Bytes available on Drive B:

DISC 4A

-BCCMPMD 04A 0K : COBOL2 OVR 14K : USQ COM 2K : Z80PCDS ASM 4K
 COBOL COM 30K : FOCAL AGM 36K : Z80ASM COM 10K : Z80SUBS ASM 12K
 COBOL REL 2K : SYS215 COM 8K : Z80DOC DOC 4K
 COBOL1 OVR 12K : SYMSG BB 12K : Z80MAIN ASM 28K
 14 Files (15 entries) using 174K of total 182K capacity
 49 Directory entries and 8K Bytes available on Drive B:

The astute among you will notice the cunningly concealed squeezed file in Disc 4A. This is only one. The funny files called BCCMPMD.* are part of the CP/M cataloging system. (See disc 1B)

A few notes on the discs : Disc 1A has two text adventure games on it, only ADVENTUR.COM works and is excellent. Most of the files on disc 1B are documented and need no comment from me. Disc 2A contains the really useful modem files, I'm told that MEX.COM or Modem Executive is excellent. Disc 2B has the latest version of the very useful Disc Utility DU-V86 plus the ASM and DOC files, ROFF4 is another modem file. Disc 3A has a few of the PD screen editors on it and the installation files. You will need to play around with these to get them working. These can be transferred to standard CP/M system discs for best operation. 3B has a large and complex CP/M Debugging system on it. Disc 4 has a few different computer languages on it (COBOL & FOCAL) plus the Z80 assemblers. You will need to send EITHER 4 discs made into double sided by physically punching holes etc., OR 7 "undoctored" standard discs. These will be formatted for you to the custom format system. The file on disc 3B is the instructions for the custom format system. If you don't already have custom format you will need to send an extra disc for the few special files needed to run it.

COM'ON FOLKS GIVE IT A GO -- CP/M CAN BE FUN

BEFORE THAT MS-DOS STUFF IT'S ALL THERE WAS. DID YOU KNOW THAT UP
TO 200,000 PEOPLE STILL USE IT ?

SCREEN PRINT FROM CP/M By Bob Curwood

The program printed out below is called PAGE80.ASM and this can be converted into a "COM" file which will enable you print out any 80 column screen from CP/M. The following notes are given to assist you to carry out the compiling step to produce the COM file. This ASM file will NOT run under basic or CP/M, it is intended to be used ONLY as follows.

Firstly type it in from the newsletter EXACTLY as it is printed and save it to a disc under CP/M and name it PAGE80.ASM . This is best done using a program like Wordstar ("N" option) or CP/M's ED.COM program or in fact any CP/M editor program. This is often the trickiest bit for a learner and so we do offer the file from our Public Domain library (send disc + return mailer plus stamps please). We can even give you the "COM" file as well if you want it, but the COM file cannot be printed in the newsletter and thats where the ASM file comes in.

Place the disc containing the file in the A: drive and make sure you have ASM.COM and LOAD.COM on the same disc. Copy these files from your system disc onto your disc above. Then type the following : ASM PAGE80 <RETURN>. the drive will turn on and do it's thing for a while, ignore the messages as long as there are no errors. If there are errors then you have probably made a typing error or something is wrong with the formatting of the disc or perhaps..... Anyway check EVERYTHING and try again. When the disc stops you will find several files on disc A, one of which is called PAGE80.HEX. (PAGE80.PRN contains your original ASM file)

This HEX file contains the 8080 machine language in INTEL "HEX" format suitable for the next step.

Next type this : LOAD PAGE80 < RETURN >. The drives turns on yet again and the computer will create a COM file from the HEX file. This COM file is the one which will execute directly by typing PAGE80 <RETURN>.

Once you have done this the screen printer program is "resident" until the next warm boot. The 80 column CP/M screen will now be dumped to the printer every time you press that normally silly "PRINT" key. Remember you can stop it by using CONTROL C (^C).

Note :Jim Oliver wrote the above text to support the following assembler file which is all Bob Curwood's work. Many Thanks Bob.

; 80 Col page print. 18:June:1988

; Bob Curwood. Christchurch Spectravideo Users Group

; This is a machine-specific program for the Spectravideo 328
 ; It relocates to a spare section of the FDOS and alters a hook
 ; called by the VDP interrupt (in the BASIC rom). The main section
 ; of the program, from START to ENDWRK is identical to the BASIC
 ; version, except it has been changed to 8080 code. This part
 ; actually runs while CP/M is switched out and therefore cannot
 ; use any of the BDOS calls.
 ;
 ; Note that the printer is connected to the PARALLEL port so anyone
 ; running CP/M with a serial printer is out of luck! No line feeds
 ; are output at the end of each line, so the printer must NOT be in
 ; condensed mode (i.e. it must print 80 chars per line)
 ; The printout must be interrupted (if necessary) with CTRL C not
 ; CTRL STOP. The computer will LOCK-UP if CTRL STOP is pressed while
 ; page print is doing it's thing !
 ; For an explanation of how the program works, see the BASIC version
 ; from last the newsletter

```

org 100h
chplpt: equ 03915h
beep:   equ 40beh
hook:   equ 0fe79h
offset: equ 0f000h
page:   equ 780h          ;730 for 23 lines only
reside: equ 0f810h       ;95 bytes free here

relocat: lxi h,start
         lxi d,reside
         lxi b,endwrk-start ;Routine is 88 bytes long
         db 0edh,0b0h      ;LDIR

alterhk: lxi h,reside
         shld hook+1
         mvi a,0c3h
         sta hook

message: mvi c,09
         lxi d,msg
         jp 0005

msg:     db 0ch,0ah,0ah,0ah
         db '
         db '
         db 0ah,0ah,0ah,0dh
         db '
         db '          The PRINT key next to the'
         db ' arrow keys is',0ah,0dh
         db '          now enabled. Abort a print-out',0ah,0dh
         db '          with CTRL C not CTRL STOP*'

start:   mvi a,8
         out 96h
         in 99h
         ani 20h
         rnz

prnton:  in 12h
         ani 1
         rnz
    
```

```

main:      lxi h,offset
           lxi d,offset+page
           mvi a,0ah
           call chplpt

loop:      di
           mvi a,0ffh
           out 58h
           mov c,m
           inr a
           out 58h
           inx h

ctrlc:     mvi a,6
           out 96h
           in 99h
           ani 2
           db 20h,noint-$-1           ;JR NZ
           mvi a,2
           out 96h
           in 99h
           ani 8
           db 28h,exit-$-1           ;JR Z

noint:     mov a,c
           cpi 5fh
           db 38h,alpha-$-1         ;JR C
           cpi 0beh
           db 30h,graphic-$-1       ;JR NC

inverse:   adi 40h
           ;This gives italics on my
           ;printer. Change adi 40h
           ;to SBI 63
           ;for normal ASCII
           ;JR

alpha:     db 18h,prtchr-$-1
           adi 20h

graphic:   db 18h,prtchr-$-1       ;JR
           nop                     ;Future SVI graphic routine
           nop                     ;goes in here

prtchr:    call chplpt
           rst 4
           db 20h,0c9h
           mvi a,0ah
           call chplpt
           call beep

exit:      ret

endwrk:    equ $

```

Disc / Tape 1 SPECTRAVIDEO

ADDER Mathematics
Music 3 note cords on 1 key
TIC Tic Tac O's & X's
BLITZ Creatures from the sky
LUNEY Space craft landing
gofh Gates of Hell
Disass Disassembler
robons Swamp game creatures
bounce Bouncing ball
OYHELL Othello
tron avoid trace
ski slalom course
aust map of Australia
world " " world
snake snake whos tail grows
poker card game
planet planet X
m/math moon maths
hang hangman
cycle motor cycle jump
hand joystick drawer

Disc / Tape 2 SPECTRAVIDEO

pub2 intro New Plymouth
z80plu memory location
galaxy space craft recovery
sub sub hunt
font font editor
bird bird logo
patter pattern
count Line 0,0,1
test SV logo V 5.0
blist lists a program
comman SV commands
datash data sheet No4
..... screen dump ripples
..... screen dump dip

Disc / Tape 3 SPECTRAVIDEO

pub 3 intro
ritema Ritemans scr. dump
diskof spin off patch
svmonf address / bytes
trek Star trek-Spiers 20K
trekin " " intro
3Dplot plot of formula
STREK Star trek Mod 22K
basit CP/M --> basic
cpmit basic--> CP/M
bisect Stanley bisection
state running a country
stripb strip spaces prog.
systra disk director
sports team draw
blackb blackboard
don 1 disassembler
budget home budgeting
h/hous haunted house game

Disc / Tape 4 SPECTRAVIDEO

pub4 intro
fred disc utility
hexloa M/Code Ld. Disass
codeco code to RAM copy
drop 1 SV logo drops
drop 2 2 SV logo drops
circle circle graphics
triang triangle graphics
saturn Saturn's rings
disco disc copy prog.
exempl basic vs M/C
yatzi Yahtzi dice game
fuel fuel consumption
shapes shapes designer

Disc / Tape 5 SPECTRAVIDEO

race Racing car
bacter Circle graphics
Sight Sight and shoot
maths Speed maths
blackJ Blachjack card game
Acesht Ace shoot sprites
Spd Spider bomb

Disc / Tape 6 SPECTRAVIDEO

SDUMP High Speed Norm/Doub Density
DUMPS High Speed Norm/Doub Density
edit Text Ed - Scroll/255 Column
CHUNG blood for edit &HC41B
ins saved instructions for edit
disass Disass 2 Memory/Disk/Tape
Biorhm Biorhythm

Shanes Shanes sprites
 B&Wrag Black & White rag
 m code machine code
 hexdum hex dump
 marque marque large print
 futvoy Futuristic Voyage
 symbol Chemical symbols
 optics Optical Illusions
 vernie Vernier caliper
 120col 120 SVI colours
 Flight flight simulator

Compat Compatibility
 Calend Calendar 1AD to 1999
 INLAY Music Cassette inlay
 INLAY2 General Purpose inlay
 stats Statistics
 graph Bar Graph up to 40 variab
 GPHlpt Prints you Fonts to print
 Format Formats an ASCII saved pr
 Z80plu Memory Peeking and Poking
 blist Lists loc of program lines
 WORDS Words puzzle

Disc / Tape 9 SPECTRAVIDEO

Alcoho Alcohol Absorption
 Othell Othello Board Game
 cup World Cup soccer game
 patien Patience Card Game
 kalid Kalidoscope Graphics
 starga Star Gate Graphics
 tunnel Tunnel Graphics
 React Reaction Time Test
 Life Life Expectancy
 Displa Displays messages
 Format Formats a program
 Mon Memory Peeking & Poking
 D1 3D sphere
 GrpM1 Graphic Mode 1 Displays
 Topsy Inverted Characters
 searc8 search disc file
 searc4 search disc file
 text80 disk file sve. ld. ed.
 text40 " " " " "
 draw& Coloured RND circle

Disc / Tape 10 SPECTRAVIDEO

Volley Volley Ball
 Saveso Save your soul
 testpa Test pattern / clock
 Miner Diamond Miner
 annie Music - Annie
 wildca Wildcat oil strike
 Sub Sub Hunt
 lady 80 column picture
 choppe Sound - Helicopter
 lotto Tattslotto Carding
 Temp Temperature Convertr.
 piano PLAY a Piano
 colors Colour and Paint
 palind Palindrome Numbers
 12 day 12 Days of Christmas
 phones Telephone sounds

Disc / Tape 7 SPECTRAVIDEO

Volcan Volcano Island (SV328)
 P of Z Planet of Zandell (SV328)
 Kings& Kings and Queens
 LORDof Lord of the Rings (SV328)
 Snaker Snake and Obstacles
 Star 1 Star lanes
 towers Towers of Hanoi puzzle
 bomb Bomb planet and land
 free f Free fall parachuting
 rescue Rescue from Eruption
 puzzle Mathematics puzzle
 Conser Conservator and Multinash
 (All games from the Australia U G)

Disc / Tape 8 SPECTRAVIDEO

SpecDr Spectra-Draw(Screen dr
 draw Screen draw
 pen Screen draw
 FontEd Font Editor
 Sound Sound Command Utility
 Sprite Sprite 8*8 Utility
 Diss Disassembler L.Dunning
 ASMZ80 Assembler D.Elson
 DDT Display
 VLINK Video Ram loader
 NumCon Number Converter Bin
 TextEd Text Editor
 polyNo Algebra Maths
 cube Cube Screen rotation
 shuttl Space Shuttle Graphic
 Data70 Font Set

Disc / Tape 11 SPECTRAVIDEO

Collis Collision Game
 Hang M Hang man word game
 Shoot Shoot game
 Mirror Mirror game
 Pacman Pacman game
 Sprite Sprite maker
 House Draw House
 Happy Happy Birthday
 S size Sprite sizes
 3D plot 3D formula plot
 Scroll Screen 2 scroll
 CHR 0 M.C. Chr display
 Variab Variables trace
 Rat Rat trap security
 print Font screen print
 Enlarge Font enlargement
 Scrswp Screen swap
 errors redefines errors
 repeat CMD Repeat & Until

Disc / Tape 12 SPECTRAVIDEO

DATA Puts memory into Data lines
 BIN RD Machine Code header reader
 Rotate Rotate name table
 Line Line command variations
 Appear Screen 1 appears instantly
 Border Border rotates
 Sprite 32 Sprites move
 Astro Astro Blaster
 Metro Metronome for music timing
 Maths Maths tester
 Blasta Blasting Duel
 Master Mastermind
 Aleyin Aleyin invasion
 Garden Garden record keeper
 prgdir Program directory file
 Colors Colour screen/border/sprite
 Sprite Sprite
 Snoopy Snoopy picture

Disc / Tape 13 SPECTRAVIDEO

Marble Numble Marble Puzzle
 cars Bathurst car race
 pengui Save the Penguin eggs game
 evasion Evasion of the ???? game
 IBM I B M Logo
 spiral Spiral graphics
 point Point graphics
 golf Golf Game 9 Holes
 airfox Airfox adventure game
 doc Doctor analyst
 sort Sorts words
 anagram Anagram's anything
 sort 2 Sorts words
 switch Bank switch 29k more ram!
 Naval Naval Combat board game
 Notes Music Tutorial

Disc / Tape 14 SPECTRAVIDEO

dump Screen Dump 11 - B. Parker
 Banner Banner - for Printers
 Sort Sort by count
 Words Hidden Word puzzle
 PDCalc Calc Spread Sheet 80 Col
 Life Life Expectancy
 G Prix Grand Prix Racing game
 GoBang Russian Roulette
 PokerM Poker Machine
 Horse Horse Racing
 Madlib Using words in diffnt docs
 label Address label
 quest Quest adventure game
 poetry Poetry Composer
 Toader Toad crossing the road

Disc / Tape 17 SPECTRAVIDEO

hexap large hexpaw prog.
 sahara German language game
 suche suche?
 gopher zap a gopher
 engulf screen 2 game
 arkra adventure game
 stock stocks & shares game
 y=f(x) function plotter
 q*bert q-bert can jump game
 file disc file utility

Disc / Tape 18 SPECTRAVIDEO

chess chess game
 vpoke video ram demo
 swp23 basic ver. of 'sweep'
 screen instructions for swp23
 pad make a sprite with pad
 gunner gunnery practise
 cavern Caverns of Venus game
 room room maze game
 bomb French bombing game
 bio3 compatibility & biory.
 baccar baccarat card game

Disc / Tape 21 SPECTRAVIDEO

IQcode	M/C file for Iquest
iplprog	Disc ipl driver
docgraph	Graph documentation
boot	Menu prog.
iqdata	Data for Iquest
docsuper	S/Artist Documentn.
enh_dem	Enhance demo
enhance	Enhance prog.
graph	Prints out docgraph
art	" " docsuper
quest	Quest adventure game
enh_doc	Enhance doc.file
graphp	graph plotter prog.
supera	Super artist prog.

Disc / Tape 22 SPECTRAVIDEO

cannon	howitzer shoot up
chess	simple chess
hang	hangman game
invasn	invasion of CH-CH
names	club list (excellent)
poker	poker game
pont	pontoon cards
sums	arithmetic
terror	shoot & dodge
yahtzee	yahtzee game
bombs	bomb the city
lstrom	key words list
revers	game of skill
stark	space invaders
pd	pd file

Disc / Tape 15 SPECTRAVIDEO

input	Aust.Users input program
play k	Play ground
morgag	Morgage calculations
scan	Disk scanner
escape	Escape frm Seralon adventure
crysta	Crystal graphics
worm	The worm which grows game
snowfl	Snow flake graphics
slide	Letter puzzle
mindbu	Mindbuster puzzles
stars	Star Graphics
Yahtze	Yahtze board Game
Boing	Action Game
1 Arm	One arm bandit
tunes	Musical tunes
Smart	A.I type question and answer
Vram	WNSV mag page 19 Sept 86
Err	Error messages and codes
Intere&Lprint	Interest & screen to prt

Disc / Tape 16 SPECTRAVIDEO

joust	Fly and joust game
tiler	Tile fast game
labyrn	Memory labyryth game
effect	sound effects library
flute	Search for Gold flute
hexal	Hexapawn game
bobo	Man gets fruit game
data77	Computer font loader
awari	Awari game
new y	Bomb New York!
countr	Countries quiz
bach	Bach music
d-kong	Donkey kong game

Disc / Tape 19 SPECTRAVIDEO

FIVE	:
PACK2	:
INV	:
MAIN	!<=>KINGS SOLOMONS MINES
PACK1	:(EXCELLENT ADVEN. GAME
THREE	:(GRAPHICS & SOUND)
triffid	Catch & dodge game
airattack	action game
bach1	bach music
tenpin	bowling tenpin
disk.dev	B.Fisk's disc util
filedir.11	Basic disc dir. util

Disc / Tape 20 SPECTRAVIDEO

SWITCH	M/C bank switch
EDITOR	Disc editor
STRIP.	Strip REM's
ARUN40	Menu prog.
3D-TIC	3D tic-tac-toe
BANK21	Bank switch
CANCEL	Used by Discat
BINCAS	M/C headers (tape)
L-FRMT	New version format
PARAJP	Parachute jump
NSWP.	Disc copy (excell)
RENAME	Used by Discat

Disc /Tape 23 SPECTRAVIDEO

catch8.8	80 column catch it game
se23	text screen to basic prog.
PROOF1	COMPUTE! proof typer
revers4	Othello game (fast)
grab-em	PAC MAN game
cup	football teams game
solitaire	card game *
monte2	monte carlo cards *
masterm	detective who done it
faceache	face drawing prog.
hikdoc	tell the time prog.*
wari2	Awari game *
prtscr	M/C text print screen
life40	Game of life
vpoke	vpoked fonts (clever)
DAIRY	diary keeper
scroll	M/C text scroll
mod.bas	graphics demo *
kaleid	kaleidoscope colors *
tennis	simple tennis game
qlott	!
conesv	!==> 3D plotters
parabo	! (graphics)
hyperb	!
plotter	3D plot of functions

* --- from COMPUTE! magazine

3D-LAB

WINDOW
DISCAT
UPDCAT
CHECK.
SUPERS
macode

3D labyrinth adv.

Windows demo
Disc Cataloguer
Updates Discat
Checks for text
Game
Menu driven M/C prog.

Disc /Tape 24 SPECTRAVIDEO

This disc was generously sent to us by Bob Curwood from Christchurch. It contains some of the best program protection schemes ever devised for SV. The challenge is up to you ; break down all the protections used AND get an excellent circuit drawing utility as well. This will enable you to draw almost any electronic symbol and to place them into a circuit drawing on screen. This can be dumped to a printer or saved to disc. There are 40 odd sprites used (resistors, capacitors, IC'S, transformers etc,etc) and you can add your own. This disc contains some high quality stuff which will keep you amused and amazed for hours.

SPECTRAVIDEO BASIC TOKENS by Jim Oliver

You will all know that commands such as PRINT, LIST, BEEP and all the others are called "reserved words" or "key words" in many different versions of Basic. Spectravideo Basic has perhaps more than most which makes our Basic so easy to use.

What you may not realise is that the computer uses a series of one or two byte "words" or "tokens" to represent these reserved words in a program in the RAM or memory. This means a considerable saving in memory space can be achieved. For example the word BEEP in hex is 42,45,45,50 but in memory it is C0 hex. There is a table in ROM (starting at &H2CB) which tells the computer what's what.

Below is a listing giving all the tokens and the reserved words from this ROM area. These have been presented before but not in token numerical order as here. This is most useful if you are 'hacking' away through a memory dump. See newsletter number 7 (January / February 1985) for a full description of this matter. Please note that there are many more reserved words, eg INTERVAL can you work out why this is and how these are saved in memory.

For the more experienced the following : The area of ROM is from &H2C8 to &H599. The first character is missing. The last character has &H7F added to it. This is followed by the token. If the token is > &H7F then the token is one byte, if the token is < &H7F then the token has &HFF added to it as a separate byte. In this case the token in ROM may be shown as FF 0E (ATN) but in memory the second byte has &H80 added to it, so ATN becomes FF 8E.

WORD	TOKEN	WORD	TOKEN	WORD	TOKEN	WORD	TOKEN
END	81	HEX\$	FF,9B	CIRCLE	BC	IMP	FC
LEFT\$	FF,81	LPOS	FF,9C	COLOR	BD	MOD	FD
FOR	82	OUT	9C	DRAW	BE		
RIGHT\$	FF,82	BIN\$	FF,9D	PAINT	BF		
MID\$	FF,83	LPRINT	9D	BEEP	C0		
NEXT	83	CINT	FF,9E	PLAY	C1		
DATA	84	LLIST	9E	PSET	C2		
SGN	FF,84	CSNG	FF,9F	PRESET	C3		
INPUT	85	CLS	9F	SOUND	C4		
INT	FF,85	CDBL	FF,A0	SCREEN	C5		
ABS	FF,86	WIDTH	A0	VPOKE	C6		
DIM	86	ELSE	A1	KEY	C7		
READ	87	FIX	FF,A1	CLICK	C8		
SGR	FF,87	STICK	FF,A2	SWITCH	C9		
LET	88	TRON	A2	MAX	CA		
RND	FF,88	STRIG	FF,A3	MON	CB		
GOTO	89	TROFF	A3	MOTOR	CC		
SIN	FF,89	PDL	FF,A4	BLOAD	CD		
LOG	FF,8A	SWAP	A4	BSAVE	CE		
RUN	8A	ERASE	A5	MDM	CF		
EXP	FF,8B	PAD	FF,A5	DIAL	D0		
IF	8B	DSKF	FF,A6	DSKO\$	D1		
COS	FF,8C	ERROR	A6	SET	D2		
RESTORE	8C	FPDS	FF,A7	NAME	D3		
GOSUB	8D	RESUME	A7	KILL	D4		
TAN	FF,8D	CVI	FF,A8	IPL	D5		
ATN	FF,8E	DELETE	A8	COPY	D6		
RETURN	8E	AUTO	A9	CMD	D7		
FRE	FF,8F	CVS	FF,A9	LOCATE	D8		
REM	8F	CVD	FF,AA	TO	D9		
INP	FF,90	RENUM	AA	THEN	DA		
STOP	90	DEFSTR	AB	TAB(	DB		
PRINT	91	EOF	FF,AB	STEP	DC		
POS	FF,91	DEFINT	AC	USR	DD		
CLEAR	92	LOC	FF,AC	FN	DE		
LEN	FF,92	DEFSNG	AD	SPC(	DF		
LIST	93	LOF	FF,AD	NOT	E0		
STR\$	FF,93	DEFDBL	AE	ERL	E1		
NEW	94	MKI\$	FF,AE	ERR	E2		
VAL	FF,94	LINE	AF	STRING\$	E3		
ASC	FF,95	MKS\$	FF,AF	USING	E4		
ON	95	MKD\$	FF,B0	INSTR	E5		
CHR\$	FF,96	OPEN	B0	VARPTR	E7		
WAIT	96	FIELD	B1	CSRLIN	E8		
DEF	97	GET	B2	ATTR\$	E9		
PEEK	FF,97	PUT	B3	DSKI\$	EA		
POKE	98	CLOSE	B4	OFF	EB		

WORD	TOKEN	WORD	TOKEN	WORD	TOKEN	WORD	TOKEN
VPEEK	FF, 98	LOAD	B5	INKEY\$	EC		
CONT	99	MERGE	B6	POINT	ED		
SPACE\$	FF, 99	FILES	B7	SPRITE	EE		
CSAVE	9A	LSET	B8	TIME	EF		
OCT\$	FF, 9A	RSET	B9	AND	F8		
CLOAD	9B	SAVE	BA	OR	F9		
LFILES	BB	XOR	FA	EQV	FB		
END	81	HEX\$	FF, 9B	CIRCLE	BC	IMP	FC
LEFT\$	FF, 81	LPOS	FF, 9C	COLOR	BD	MOD	FD
FOR	82	OUT	9C	DRAW	BE		
RIGHT\$	FF, 82	BIN\$	FF, 9D	PAINT	BF		
MID\$	FF, 83	LPRINT	9D	BEEP	C0		
NEXT	83	CINT	FF, 9E	PLAY	C1		
DATA	84	LLIST	9E	PSET	C2		
SGN	FF, 84	CSNG	FF, 9F	PRESET	C3		
INPUT	85	CLS	9F	SOUND	C4		
INT	FF, 85	CDBL	FF, A0	SCREEN	C5		
ABS	FF, 86	WIDTH	A0	VPOKE	C6		
DIM	86	ELSE	A1	KEY	C7		
READ	87	FIX	FF, A1	CLICK	C8		
SQR	FF, 87	STICK	FF, A2	SWITCH	C9		
LET	88	TRON	A2	MAX	CA		
RND	FF, 88	STRIG	FF, A3	MON	CB		
GOTO	89	TROFF	A3	MOTOR	CC		
SIN	FF, 89	PDL	FF, A4	BLOAD	CD		
LOG	FF, 8A	SWAP	A4	BSAVE	CE		
RUN	8A	ERASE	A5	MDM	CF		
EXP	FF, 8B	PAD	FF, A5	DIAL	D0		
IF	8B	DSKF	FF, A6	DSKO\$	D1		
COS	FF, 8C	ERROR	A6	SET	D2		
RESTORE	8C	FPOS	FF, A7	NAME	D3		
GOSUB	8D	RESUME	A7	KILL	D4		
TAN	FF, 8D	CVI	FF, A8	IPL	D5		
ATN	FF, 8E	DELETE	A8	COPY	D6		
RETURN	8E	AUTO	A9	CMD	D7		
FRE	FF, 8F	CVS	FF, A9	LOCATE	D8		
REM	8F	CVD	FF, AA	TO	D9		
INP	FF, 90	RENUM	AA	THEN	DA		
STOP	90	DEFSTR	AB	TAB(	DB		
PRINT	91	EOF	FF, AB	STEP	DC		
POS	FF, 91	DEFINT	AC	USR	DD		
CLEAR	92	LOC	FF, AC	FN	DE		
LEN	FF, 92	DEFSNG	AD	SPC(	DF		
LIST	93	LOF	FF, AD	NOT	E0		
STR\$	FF, 93	DEFDBL	AE	ERL	E1		
NEW	94	MKI\$	FF, AE	ERR	E2		
VAL	FF, 94	LINE	AF	STRING\$	E3		
ASC	FF, 95	MKS\$	FF, AF	USING	E4		
ON	95	MKD\$	FF, B0	INSTR	E5		
CHR\$	FF, 96	OPEN	B0	VARPTR	E7		
WAIT	96	FIELD	B1	CSRLIN	E8		
DEF	97	GET	B2	ATTR\$	E9		
PEEK	FF, 97	PUT	B3	DSKI\$	EA		
POKE	98	CLOSE	B4	OFF	EB		
VPEEK	FF, 98	LOAD	B5	INKEY\$	EC		
CONT	99	MERGE	B6	POINT	ED		



WORD	TOKEN	WORD	TOKEN	WORD	TOKEN	WORD	TOKEN
SPACE\$	FF,99	FILES	B7	SPRITE	EE		
CSAVE	9A	LSET	B8	TIME	EF		
OCT\$	FF,9A	RSET	B9	AND	F8		
CLOAD	9B	SAVE	BA	OR	F9		
LFILES	BB	XOR	FA	EQV	FB		